

unsupervised feature learning. DL is also known as a multi-layered neural network. Just like the human brain works on neural networks, DL uses the multi-layered networks to process data and make decisions. As the name indicates, it helps in reading the hidden meaning deep inside the data.

Data science (DS)

Data science is about understanding and visualising the data to derive meaning out of it. Since there is a very thin line between the skills set and responsibilities of data scientists and data engineers, people get confused about their roles. Data engineers are focused on building infrastructure and architecture for data generation while data scientists focus on the advanced mathematics and statistical analysis of that generated data. It overlaps with what AI does and subsequently with ML (refer to Figure 1 to understand this better).

These four branches all need various mathematical tools and methods like probability, statistics, linear algebra, numerical optimisation and computer programming.

Now that we know what artificial intelligence is and how it is related with machine learning, let's try to understand a well-known application of AI, called a chatbot.

Have you ever wondered how you get an instant reply from Priya, Pankaj, Alex or John while seeking help on any banking, shopping or hotel service portal?

Initially, when I was unaware of machine learning, I used to wonder how these banks managed to hire people that resolved customer queries instantly over a chat. But ML amazed me with its magic and chatbots are one among its many applications.

Chatbots

This is defined as "A computer program (based on artificial intelligence) designed to simulate conversation with human users, especially over the Internet."

A chatbot is an automated assistant that communicates with us through text messages or voice commands. It is a virtual companion that integrates into websites, applications or instant messengers and helps entrepreneurs to get closer to customers.

Types of chatbots

Depending upon the business case, we can classify chatbots into two large categories.

A **simple chatbot** works based on pre-defined keywords and a set of rules. These rules will select an appropriate answer for a given asked question. For example, we can design a chatbot to:

- Respond to FAQs
- Check hotel room availability and bookings
- Handle flight enquiries

Smart chatbots rely on machine learning methods, wherein all the previous chats are recorded for later processing so that they get a more accurate response for every new

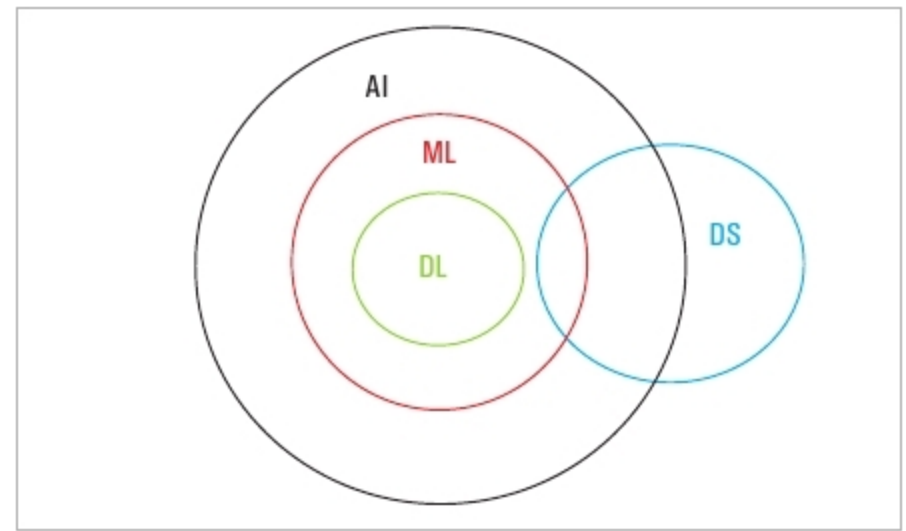


Figure 1: Venn diagram for AI and its components

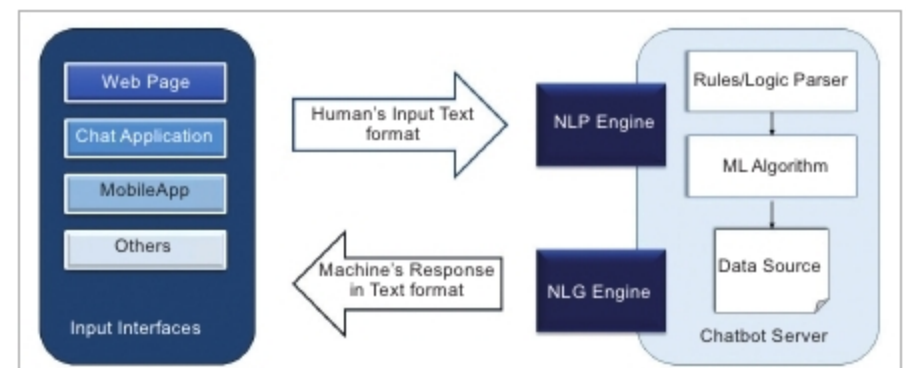


Figure 2: Structure of a chatbot

query. In this case, results are generated dynamically based on algorithms that can be applied on the data set that is specific to the problem being dealt with. Such bots can:

- Suggest a food menu based on one's previous orders.
- Act as a daily fitness tracker and provide suggestions based on current health. Extending this to the next level, a smart bot can search for nearby hospitals and also book an appointment for you.

Google Home and Amazon Alexa are the best examples of smart chatbots; they are multi-lingual and work with voice commands rather than text messages.

Structure of a chatbot

A chatbot is often described as one of the most advanced and promising interactions between humans and machines. Chatbots simulate the conversation (mostly text messages) with the user in natural language. The best thing about chatbots is that they can be integrated with any interface like in social networks, smartphone applications, messengers, etc.

Natural language processing is used to convert text input to process the query and pass it on to the logic parser, where it extracts the data/knowledge based on the ML algorithm. Once the result is generated, it is passed to the natural language generator (NLG) engine to convert the machine's output into human readable format (text), after which it is fetched by the frontend interface and presented to the user. Refer Figure 2 for pictorial representation.

Designing one's own chatbot

There are several ways to design and implement a chatbot.